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Vehicle detection using three-axis AMR sensors deployed along travel lane markings

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Abstract: A vehicle detection method is developed based on three-axis anisotropic magneto resistive (AMR) sensors along travel lane markings. The method integrates multiple algorithms and uses different geomagnetic waveforms which are disturbed by vehicles passing a single AMR sensor. This method comprehensively analyses *X*-, *Y*-, and *Z*-axis information and applies a double-window algorithm to extract a single vehicle waveform. The vehicle mixed algorithm (VMA) is developed to differentiate vehicles driving by the AMR sensor simultaneously and determine vehicle flow rates. In addition, the vehicle motion-state discrimination algorithm (VMSDA) is developed to distinguish the vehicle operating status (i.e. driving on the left lane, the lane line, or the right lane). The field experimental tests verified the effectiveness of the both algorithms. Results indicate that the average accuracy rates of VMA and VMSDA can, respectively, be up to 98.0 and 96.4%.

1 Introduction

Traffic information acquisitions are an important foundation for supporting intelligent traffic operations, and are a basic requirement for the successful implementation of intelligent transportation systems in urban areas [1]. Geomagnetic sensor is superior because of its unique advantages, such as its robustness not being influenced by climate and weather conditions, its low costs, small volumes, convenient layouts, and suitability for wireless communication [2].

The traditional geomagnetic detector is usually placed on the roadside or at the center of lanes to collect traffic measurements, which provide a limited sample dataset. Therefore, it can only analyse the vehicle motion states on the current lanes, and is unable to fully extrapolate to real-time traffic conditions. The traditional research method of vehicle detection depends on two or more geomagnetic sensors [3], thus resulting in the significant increase in vehicle detection costs. Moreover, the existing methods mostly applied single-axis information from geomagnetic sensors, giving a low-precision resolution.

This paper puts forth a vehicle detection method based on an anisotropic magneto resistive (AMR) sensor deployed along travel lane markings, which synthesises information from the three *X*, *Y*, *Z* axes to obtain vehicle driving information from two adjacent lanes, thus comprehensively analyses the current traffic conditions. In addition, the method uses a single sensor node instead of double nodes to realise the detection of traffic flow parameters, which increases accuracy, reduces the installation and maintenance costs, saves resources, and reduces layout complexity. To collect traffic information and expand the functions of the geomagnetic sensors, the vehicle detection method differentiates vehicles driving by the AMR sensor and distinguishes the position of the vehicle (driving on the left lane, the lane line, or the right lane).

2 The detection principle of AMR sensor

The Earth's magnetic field intensity is in the range of 0.05–0.06 mT on average [4]. If there is no external disturbance, the magnetic field can be regarded constant. When the vehicle enters the magnetic field, the distribution of the magnetic field will change, forming a disturbance, as shown in Fig. 1 [5]. Analysis of the perturbation of the magnetic field allows the determination of whether a vehicle has passed or not.

AMR sensors use the anisotropic magneto-resistive effect of conductor materials. The resistance of the nickel iron magnetic alloy has a function related to the angle (θ) between bias current (I) and magnetic field vector (M) [6]. It changes with the magnetic field vector. AMR has a high sensitivity and triggers a response from a small disturbance, thus detecting the changes of the geomagnetic field.

The approach places the AMR sensor on the lane line. Define the *X*-axis of the AMR sensor as the opposite driving direction set originally of driveways, the *Y*-axis as the horizontal component perpendicular to the driving direction, and the *Z*-axis as the vertical component perpendicular to the *X*-axis and the *Y*-axis, as shown in Fig. 2.

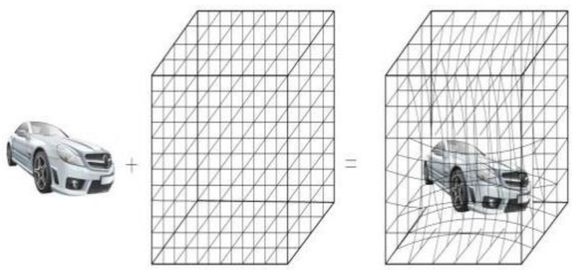

Fig. 1 *Geomagnetic field changes disturbed by vehicle*

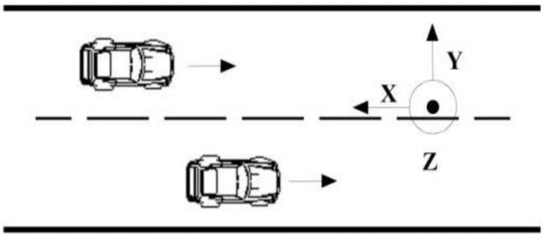

Fig. 2 *Installation layout of AMR sensors*

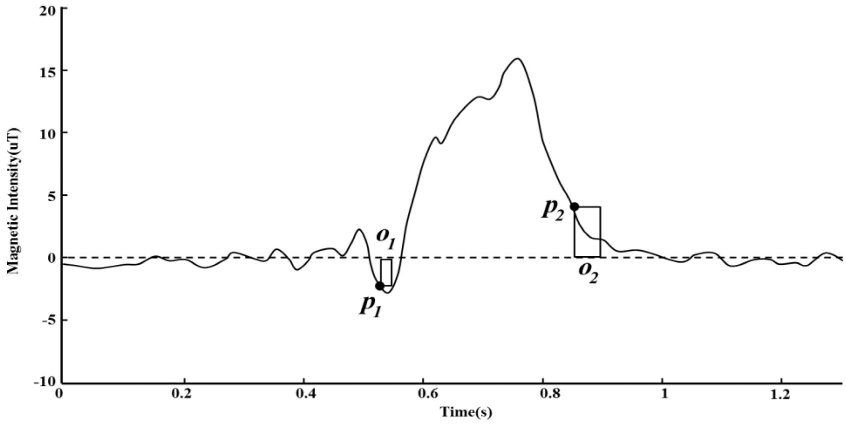

Fig. 3 *The double-window algorithm*

3 Extraction of vehicle waveform based on double-window algorithm

The data from vehicle magnetic detection is a time series of waveforms. To analyse the signal data, we define the time series as follows. The time series is a range of collected data distributed in chronological order with equal intervals, which is recorded as $X = \{X_1, X_2, \cdots, X_s\}$. Among them, $X_i = (t_i, b_i)$ is the state point of the time series, and b_i is the value of the time series at moment t_i, which is numeric data, and s is the length of the time series. For the fixed sampling frequency, time strictly increases in the time series. Namely, if $i < j$, so $t_i < t_j$. Since the time-series data has equal intervals, it can be directly represented as $X = \{x_1, x_2, \cdots, x_s\}$, where x_i is the value of the time series at moment i.

When there are no vehicles in the detection range of the AMR sensor, the detected value fluctuates with a small range of changes, which is referred to as the base value [7]. As the external environment changes, the base value varies, so a dynamic base value algorithm is needed to ensure that the vehicle waveform is extracted completely and correctly. In this paper, a large-amount average algorithm is adopted as follows. First, a large amount of data is obtained from the time-series $X = \{x_1, x_2, \cdots, x_s\}$, where x_i is the value of the time series at moment i. Then the average value of x_i is calculated, denoted as $\bar{x}$ is the initial base value D, and the time series is corrected according to D. The result is $X_{\text{new}} = \{(x_1 - \bar{x}), (x_2 - \bar{x}), \ldots, (x_s - \bar{x})\}$.

This method utilises a double-window algorithm (as shown in Fig. 3) to extract the waveform of each vehicle as follows:

(1) Initialise the size of two windows, including the vehicle-adjacency-discrimination window o_1 and the vehicle-departure-discrimination window o_2.

(2) Start the vehicle-adjacency-discrimination window o_1.

(3) Input the time series in turn. If the sampling data is on the outside of o_1, turn to (4), and record the vehicle-adjacency point as p_1, or turn to (2).

(4) Start the vehicle-departure-discrimination window o_2.

(5) Input the time series in turn. If the sampling data is inside of o_2, record the vehicle-departure point as p_2. At the same time, the vehicle flow adds one and records the waveform between p_1 and p_2 as the vehicle waveform, return to (2) to continue detecting the next vehicle, or turn to (4).

The double-window algorithm can get the complete waveform of a single vehicle, and record the sampling points of the vehicle adjacency and departure, which could provide the basic vehicle waveform data for calculating traffic flow, vehicle speed, vehicle length, time headway, vehicle type, and other traffic attributes.

4 Vehicle detection algorithm

4.1 Mathematical model of geomagnetic disturbed by a vehicle

The mathematical model of geomagnetic disturbance by a vehicle is shown in Fig. 4 [8]. Vehicles consist of iron metals, so it can produce an abnormal magnetic zone around them and cause the Earth's magnetic flux lines to bend, which changes the magnitude of magnetic induction intensity around a vehicle. The mathematical model describing magnetic signals produced by the vehicle can be simplified as a bipolar magnet, of which the magnetic moment m is in the center of the vehicle, parallel to the magnetic field, and generates magnetic components as B_X, B_Y, B_Z.

$$\begin{cases} B_X = \dfrac{\mu_0}{4\pi}\dfrac{m_x\left(2x^2 - y^2 - z^2\right) + 3m_yxy + 3m_zxz}{r^5} \\ B_Y = \dfrac{\mu_0}{4\pi}\dfrac{m_y\left(2y^2 - x^2 - z^2\right) + 3m_xxy + 3m_zyz}{r^5} \\ B_Z = \dfrac{\mu_0}{4\pi}\dfrac{m_z\left(2z^2 - x^2 - y^2\right) + 3m_xxz + 3m_yyz}{r^5} \end{cases} \tag{1}$$

where μ_0 is the permeability of free space and m_x, m_y, m_z are the magnetic moments of directions X, Y, Z, respectively, and r is the distance from the bipolar magnet to the observation point.

The expression of the magnetic induction intensity under this circumstance is as follows [9]:

$$\overline{B} = \frac{\mu_0}{4\pi}\left[\frac{3\left(\overline{IS}\overline{r}\right)\overline{r} - \left(\overline{r}\overline{r}\right)\overline{IS}}{\left(\overline{r}\overline{r}\right)^{5/2}}\right] \tag{2}$$

In (2), $\overline{I}$ is the current intensity and $\overline{S}$ is the coil area.

Table 1 Test results of VMA

Time quantum	Actual vehicles	Computational vehicles	Accuracy rate, %	Average accuracy rate, %
7:30–8:30	1128	1096	97.16	98.0
8:30–9:30	1304	1260	96.63	
9:30–10:30	612	604	98.69	
10:30–11:30	696	696	100.00	
11:30–12:30	516	524	98.45	
12:30–13:30	556	560	99.28	
13:30–14:30	412	416	99.03	
14:30–15:30	428	420	98.13	
15:30–16:30	456	460	99.12	
16:30–17:30	380	368	96.84	
17:30–18:30	1064	1028	96.62	
18:30–19:30	1204	1248	96.35	
19:30–20:30	1108	1080	97.47	
20:30–21:30	808	832	97.03	
21:30–22::30	872	880	99.08	

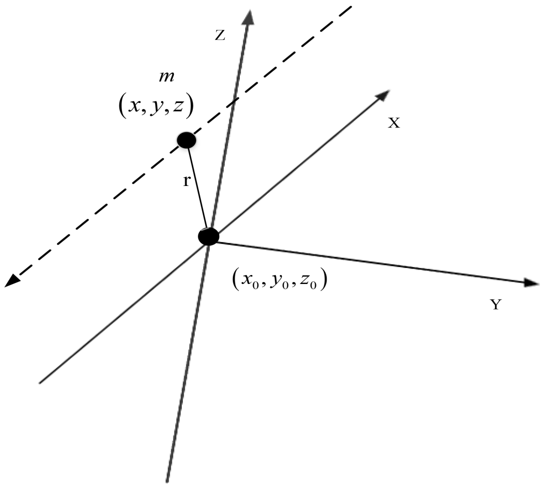

Fig. 4 *Model diagram of geomagnetic field disturbed by vehicles*

4.2 Vehicle mixed algorithm

Since the AMR sensor is placed on lane markings, it is possible for two vehicles, one on the left side of it and the other one on the right side, passing the sensor at the same time. In this case, the double-window algorithm is unable to differentiate two vehicles, which degrades the accuracy rate of vehicle detection. Therefore, we apply the vehicle mixed algorithm (VMA) to distinguish two vehicles on the left and right lanes driving through the sensor's detection area simultaneously. So that we can estimate the traffic flow more accurately and improve the detection accuracy.

By (2), the perturbation of the magnetic field produced by vehicles is proportional to $\overline{S}$, so when two vehicles pass the AMR sensor at the same time, the geomagnetic perturbation is stronger than that generated from a single vehicle passing the sensor. It means that the detected magnetic signal is larger in amplitude than that from one vehicle, as shown in Fig. 5.

In Figs. 5*a–c*, when two vehicles pass the AMR sensor at the same time, the signal fluctuation amplitude of the *X*-axis and the *Y*-axis is bigger than that when a single vehicle passes the sensor. In addition, since the *Z*-axis represents the vertical component, which is perpendicular to the horizontal plane, only when the vehicle drives over the sensor can the *Z*-axis signal have a large fluctuation.

According to this feature, we define the extracted single vehicle waveform as a vehicle sequence, the number of the initial vehicle sequence as vehicle flow N_v, the final number of vehicles as N, the value of the X, Y, Z axes of the vehicle sequence, respectively, as n_x,, n_y, n_z and the threshold of the X, Y, Z axes, respectively, as D_x, D_y, D_z. VMA is as follows (as shown in Fig. 6):

(1) Initialise N_v, D_x, D_y, D_z.

(2) Input vehicle sequence in turn.

(3) If max $(|n_z|) < D_z$, turn to (4), or turn to (2).

(4) If max $(|n_x|) > D_x$, turn to (5), or turn to (2).

(5) If max $(|n_y|) > D_y$, $N = N_v + 1$. Namely, the vehicle flow adds one or turn to (2).

4.3 Vehicle motion-state discrimination algorithm (VMSDA)

In terms of AMR sensor installation layout, the positive direction of the *Y*-axis should point to the left lane and the negative one should point to the right lane. Furthermore, the magnetic intensity B is inversely proportional to the distance r from the vehicle to the sensor by (1). That is, the closer the vehicle away from the sensor, the larger the magnetic intensity. So when the vehicle on the left lane passes the sensor, cutting the geomagnetic field vertically, the detected positive-direction signal of the *Y*-axis has large fluctuations. Similarly, when the vehicle on the right lane passes the sensor, the detected negative-direction signal of the *Y*-axis has large fluctuations. The detected geomagnetic signal is shown in Fig. 7.

From Figs. 7*a–c*, when the vehicle on the left lane passes the sensor, the signal amplitude of the *Y*-axis is mostly above zero, while the signal amplitude of the *Y*-axis that belongs to the vehicle on the right lane is mostly below zero, and the signal of the *Z*-axis that belongs to the vehicle on the lane line has large fluctuations. The VMSDA distinguishes the vehicle's operating status (e.g. driving on the left lane, the lane markings, and the right lane), which can fully distinguish the traffic state. We define the

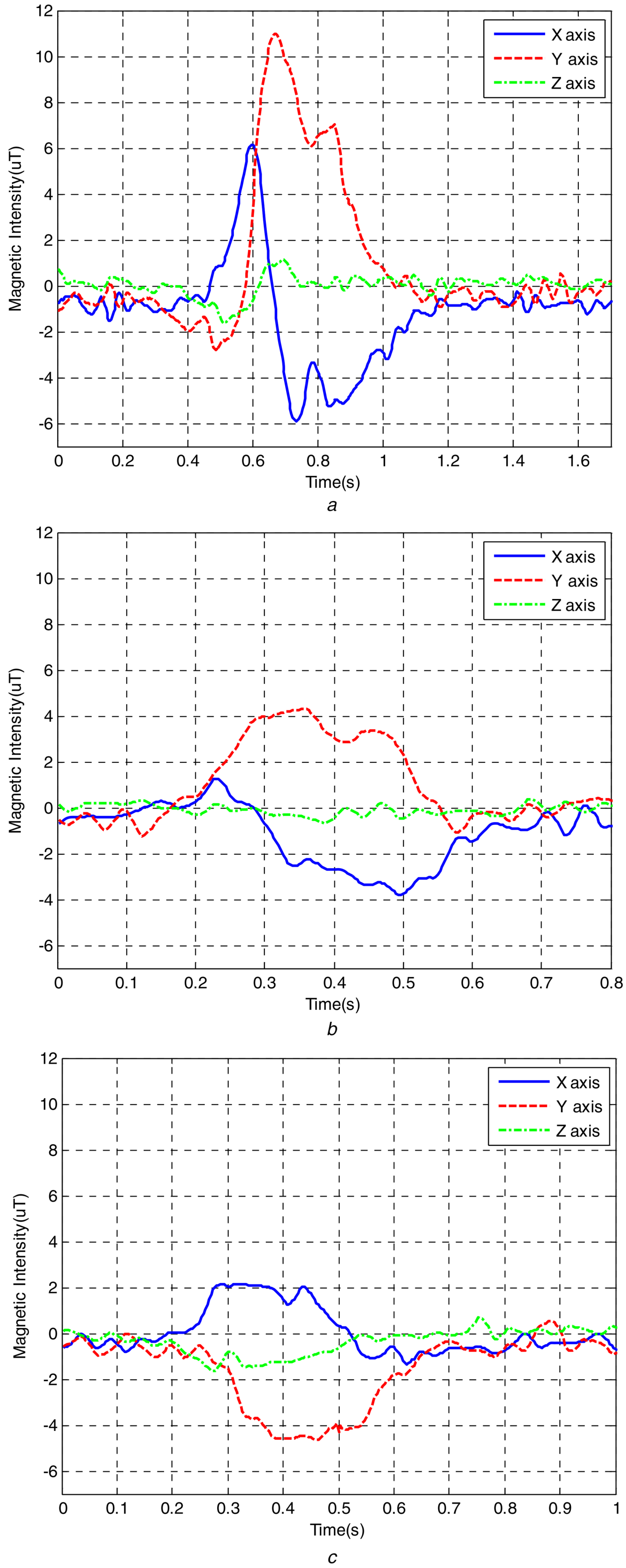

Fig. 5 *Geomagnetic signal by vehicles with different numbers*
(a) Waveform diagram of two vehicles on two adjacent lanes (one in each lane) passing a sensor simultaneously, ***(b)*** Waveform diagram of one single vehicle passing the sensor on the left-side lane. ***(c)*** Waveform diagram of one single vehicle passing the sensor on the right-side lane

thresholds of Y-axis and Z-axis, respectively, as $G_y(G_y < D_y)$, G_z, the number of vehicles driving on the left lane, the lane marking, and the right lane, respectively, as num_l, num_c, num_r, the number of vehicles passing the sensor at the same time as num_d, the final number of vehicles driving on the left lane and the right lane, respectively, as num_{lf}, num_{rf}. VMSDA is as follows (as shown in Fig. 8):

(1) Initialise G_y, G_z and num_l, num_c, num_r, num_d.

(2) Input vehicle sequences in turn, not including those when two vehicles pass together.
(3) If max $(|n_z|) < G_z$, turn to (4), or num_c adds one, turn to (2).
(4) If there is continuous $n(n > 100)$ groups $n_y < G_y$, num_r adds one, then turn to (2). Or if num_l adds one, turn to (2).
(5) Finally $\text{num}_{rf} = \text{num}_r + \text{num}_d/2$, $\text{num}_{lf} = \text{num}_l + \text{num}_d/2$.

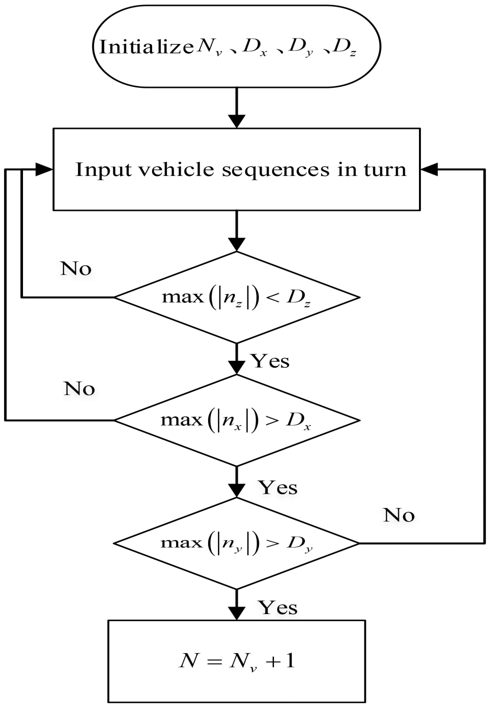

Fig. 6 *Flowchart of VMA*

5 Experimental tests and discussions

Our study utilised the sensor entitled smart road stud (SRS), which was developed independently by our team, to collect data. The SRS integrates advanced micro-sensing technology, micro-power-consumption processor technology, low-power wireless communication technology, and high-efficiency-solar-energy collection and management technology, including the energy collection unit, energy storage unit, energy management unit, lighting tip unit, magnetic sensing unit, and low-power wireless communication and control unit. The magnetic sensing unit uses BMM150, a three-axis AMR sensor. The resolution of this sensor can reach 0.3 μT, and the sampling rate is 300 Hz.

The smart road stud was deployed on the lane marking of the southbound travel lane located at the intersection between the Songshan Road and Huaihe Road in the Nangang district in the City of Harbin, China. The traffic volume is lighter and the vehicle speed is 30–40 km/h. The field experiment test conducted as shown in Fig. 9. Road data was received via a PC, and a video recording was obtained using a camera. Ground-truth vehicle counts were obtained from the video recordings. The data sample used to verify the algorithms was collected from 7 : 00–23 : 00pm on November 26, 2014. Tables 1 and 2, respectively, show the test results performed by VMA and VMSDA.

In Tables 1 and 2, the detection accuracy of the VMA and VMSDA, respectively, achieve 98.0 and 96.4%. The result shows that the feasibility and applicability of both algorithms is excellent and worth pursuing. However, the paper only analysed the situation of low speed, research of vehicle detection at a variety of sites is the next work for the follow up paper.

6 Conclusions

We applied an AMR sensor deployed on the lane markings to get the real-time vehicle waveform information. Obtaining a single vehicle waveform is suitable for vehicle flow measurements and motion-state discrimination of urban traffic by a double-window algorithm. The proposed vehicle mixed algorithm (VMA) distinguishes between two vehicles passing the sensor at the same time. It synthesises the vehicle information of the three axes, *X*, *Y*, and *Z*, of the AMR sensor, which improves the accuracy rates of vehicle detection. Another algorithm developed in the paper is the VMSDA. It synthesises the vehicle information of the *Y*-axis and the *Z*-axis and allows for the discrimination of a vehicle's operating

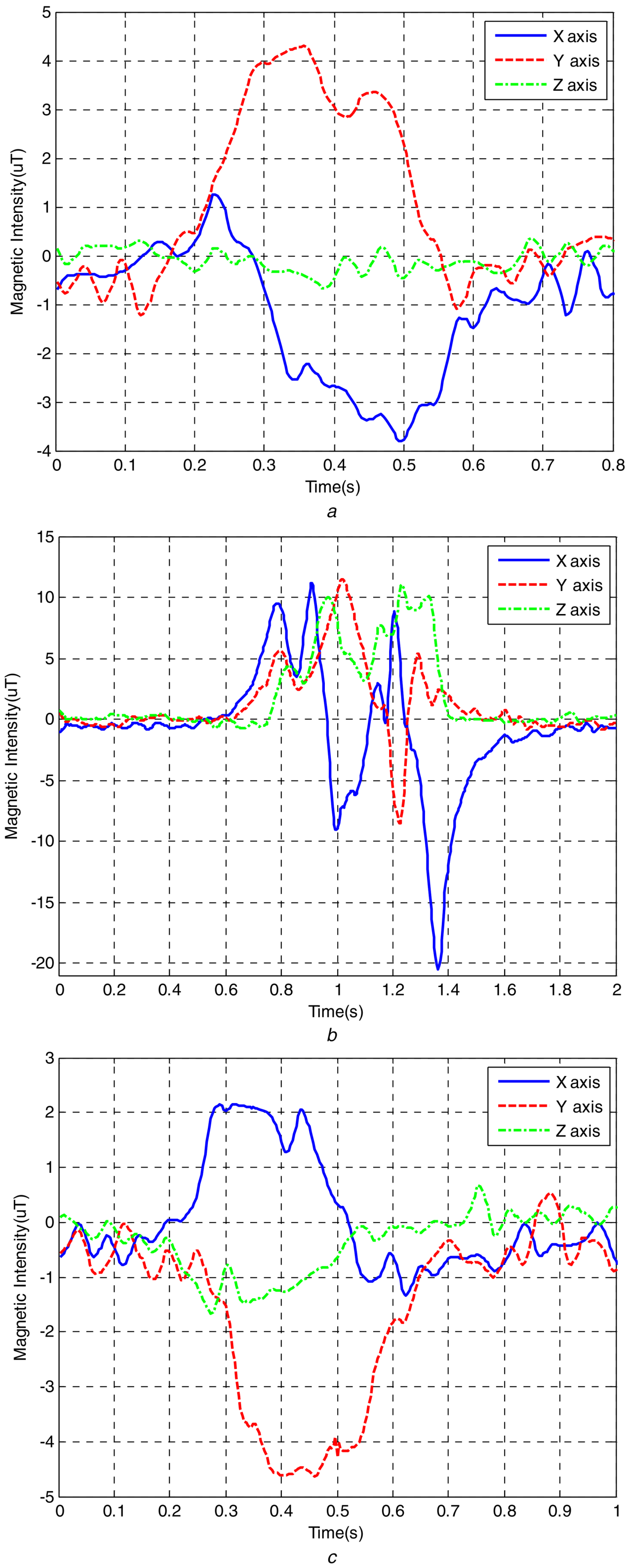

Fig. 7 *Geomagnetic signal by vehicles on different lanes*
(a) Waveform profile of vehicle driving on the left lane. ***(b)*** Waveform profile of vehicle driving on the lane markings. ***(c)*** Waveform profile of vehicle driving on the right lane

status (driving on the left lane, the lane marking, or the right lane), with high detection accuracy.

Compared with traditional vehicle detection, the study deployed the AMR sensor on the lane markings, utilised the single node, synthesised three-axis vehicle information of the sensor, and realised the vehicle flow collection and the vehicle's motion-state identification. The further research effort is suggested collecting

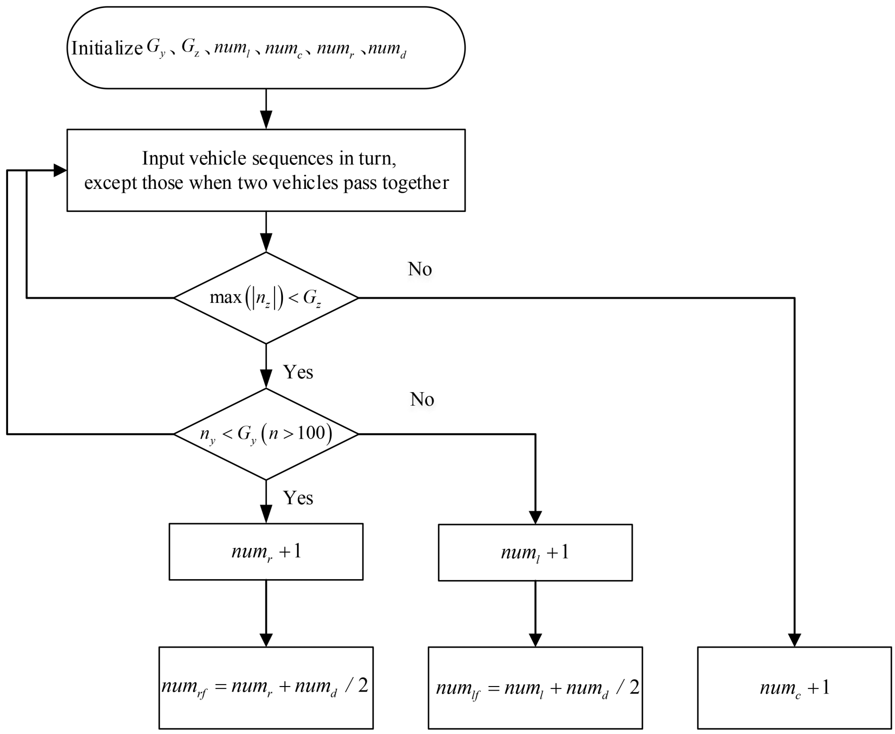

Fig. 8 *Flowchart of VMSDA*

Fig. 9 *Image of field tests*

Table 2 Test results of VMSDA

Time quantum	Actual vehicles			Computational vehicles			Accuracy rate			Total accuracy rate, %	Average accuracy rate, %
	Left Lane	Lane Line	Right Lane	Left Lane	Lane Line	Right Lane	Left Lane, %	Lane Line, %	Right Lane, %		
7:30–8:30	612	140	376	636	136	324	96.08	97.14	86.17	93.13	96.4
8:30–9:30	736	184	384	700	196	364	95.11	93.48	94.79	94.46	
9:30–10:30	292	44	276	300	40	264	97.26	90.91	95.65	94.61	
10:30–11:30	376	64	256	380	64	252	98.94	100.00	98.44	99.12	
11:30–12:30	216	28	272	220	28	276	98.15	100.00	98.53	98.89	
12:30–13:30	312	24	220	312	24	224	100.00	100.00	98.18	99.39	
13:30–14:30	172	60	180	180	60	176	95.35	100.00	97.78	97.71	
14:30–15:30	228	32	168	228	32	160	100.00	100.00	95.24	98.41	
15:30–16:30	176	60	220	184	64	212	95.45	93.33	96.36	95.05	
16:30–17:30	152	16	212	144	16	208	94.74	100.00	98.11	97.62	
17:30–18:30	588	116	360	580	120	328	98.64	96.55	91.11	95.43	
18:30–19:30	624	136	444	648	128	472	96.15	94.12	93.69	94.66	
19:30–20:30	496	80	532	480	88	512	96.77	90.00	96.24	94.34	
20:30–21:30	328	48	432	340	44	448	96.34	91.67	96.30	94.77	
21:30–22::30	420	68	384	424	72	384	99.05	94.12	100.00	97.72	

additional sample data for vehicle classification under different weather conditions.

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